# Robot Localization for PR2 Using Kalman Filter and Particle Filter

Guangxuan Xu
University of Michigan
Department of Biomedical Engineering
Department of Robotics
eggsyxu@umich.edu

**Abstract**

This project investigates the problem of robot localization for a PR2 robot navigating in an environment with obstacles. A simple location sensor is implemented in PyBullet to provide noisy observations of the robot's true position. Given a predefined and nontrivial navigation path, the robot's pose is estimated online using two probabilistic state estimation methods: a Kalman filter and a particle filter.

The sensing noise, action noise, and algorithmic parameters are carefully tuned to ensure that the localization problem is sufficiently challenging while remaining solvable. The performance of the two approaches is evaluated and compared across several representative scenarios in terms of estimation accuracy and robustness. In particular, a scenario is constructed in which the Kalman filter fails to produce a reasonable estimate—such as yielding a mean state that lies inside an obstacle—while the particle filter is still able to maintain a valid and physically plausible belief over the robot's position.

**Keywords:** Robot Localization, Kalman Filter, Particle Filter, Probabilistic State Estimation, PyBullet

## 1 Introduction

Accurate localization is a fundamental capability for mobile robots operating in real-world environments. In the presence of sensor noise, motion uncertainty, and obstacles, estimating the robot's position becomes a nontrivial probabilistic inference problem. Reliable localization is essential for tasks such as navigation, obstacle avoidance, and autonomous decision making, and therefore serves as a core component of many robotic systems [1, 2].

Classical probabilistic state estimation methods, such as the Kalman filter, have been widely applied to robot localization due to their computational efficiency and analytical tractability [3]. However, the Kalman filter relies on linear system dynamics and Gaussian noise assumptions, which may not hold in complex or ambiguous environments. When the underlying belief distribution becomes multimodal or highly nonlinear, the Gaussian approximation inherent to the Kalman filter can lead to poor or even physically invalid estimates [4, 5].

Particle filters provide an alternative framework by representing the robot's belief as a set of weighted samples, allowing them to capture non-Gaussian and multimodal distributions [6]. This flexibility makes particle filters particularly suitable for localization problems involving perceptual ambiguity, sparse sensing, or obstacle-rich environments, albeit at the cost of increased computational complexity [7].

This project focuses on the localization problem for a PR2 robot navigating in an obstacle-filled environment simulated in PyBullet. A simple location sensor is implemented to provide noisy measurements of the robot's true pose, reflecting uncertainty in practical sensing systems. As the robot executes a predefined and nontrivial path, its position is estimated using both a Kalman filter and a particle filter. Through a series of experiments with varying noise levels and environmental configurations, this project compares the accuracy and robustness of the two methods. In particular, a deliberately challenging scenario is constructed in which the Kalman filter produces an unreasonable estimate, such as a mean position located inside an obstacle, while the particle filter is still able to maintain a physically valid localization result [8].

# 2 Implementation

This section describes the implementation of the localization system used for the PR2 robot, including the motion model, sensor model, and the two state estimation algorithms: the Kalman filter (KF) and the particle filter (PF). The entire system is implemented and evaluated in a PyBullet simulation environment.

## 2.1 Motion Model

The robot state is defined as a two-dimensional position vector

$$x_k = [\, x_k,\ y_k \,]^\top,$$

representing the planar position of the PR2 base. The control input

$$u_k = [\, \Delta x_k,\ \Delta y_k \,]^\top$$

corresponds to the commanded incremental displacement between two consecutive time steps.

The motion model is given by

$$x_{k+1} = x_k + u_k + w_k, \tag{1}$$

where the process noise

$$w_k \sim \mathcal{N}(0, Q)$$

models uncertainty in the robot motion. In the implementation, the covariance matrix is chosen as

$$Q = \sigma_m^2 I_2,$$

with $\sigma_m = 0.02$. This noise is injected directly into the simulated true state to reflect actuation uncertainty.

## 2.2 Sensor Model

A simple location sensor is implemented to provide noisy measurements of the robot position. The observation model is defined as

$$z_k = x_k + v_k, \tag{2}$$

where

$$v_k \sim \mathcal{N}(0, R)$$

represents measurement noise. The sensor noise covariance is

$$R = \sigma_s^2 I_2,$$

with $\sigma_s = 0.08$. This model simulates imperfect localization sensors such as GPS-like measurements or external tracking systems.

### 2.3 Kalman Filter

The Kalman filter maintains a Gaussian belief over the robot state, represented by a mean estimate $\hat{x}_k$ and covariance $P_k$. Given the linear motion and sensor models, the Kalman filter recursion consists of a prediction step and an update step.

**Prediction**

$$\hat{x}_{k|k-1} = \hat{x}_{k-1|k-1} + u_k, \tag{3}$$
$$P_{k|k-1} = P_{k-1|k-1} + Q. \tag{4}$$

**Update**

$$K_k = P_{k|k-1}(P_{k|k-1} + R)^{-1}, \tag{5}$$
$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - \hat{x}_{k|k-1}), \tag{6}$$
$$P_{k|k} = (I - K_k)P_{k|k-1}. \tag{7}$$

The Kalman filter estimate is visualized in red during the simulation.

### 2.4 Particle Filter

The particle filter represents the belief distribution using a set of $N$ weighted particles

$$\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^{N}.$$

In this implementation, $N = 800$ particles are used.

**Initialization** Particles are initialized around the initial robot position using a Gaussian distribution with standard deviation 0.35, ensuring a reasonable initial spread while maintaining convergence stability.

**Prediction** Each particle is propagated using the same motion model as the true system:

$$x_{k+1}^{(i)} = x_k^{(i)} + u_k + w_k^{(i)}, \tag{8}$$

where $w_k^{(i)} \sim \mathcal{N}(0, Q)$.

**Update** Particle weights are updated according to the likelihood of the observation:

$$w_k^{(i)} \propto \exp\left(-\frac{1}{2}(z_k - x_k^{(i)})^\top R^{-1}(z_k - x_k^{(i)})\right). \tag{9}$$

Weights are normalized after the update, and resampling is performed internally to mitigate particle degeneracy.

**Estimation** The particle filter state estimate is computed as the weighted mean of all particles:

$$\hat{x}_k = \sum_{i=1}^{N} w_k^{(i)} x_k^{(i)}. \tag{10}$$

The particle filter estimate is visualized in green during the simulation.

### 2.5 Simulation Setup and Visualization

The PR2 robot is simulated in a PyBullet environment containing static obstacles, as illustrated in Fig. 1. A predefined two-segment trajectory is used to generate motion commands, resulting in a change in direction that introduces ambiguity and challenges the estimation algorithms, as shown in Fig. 2.

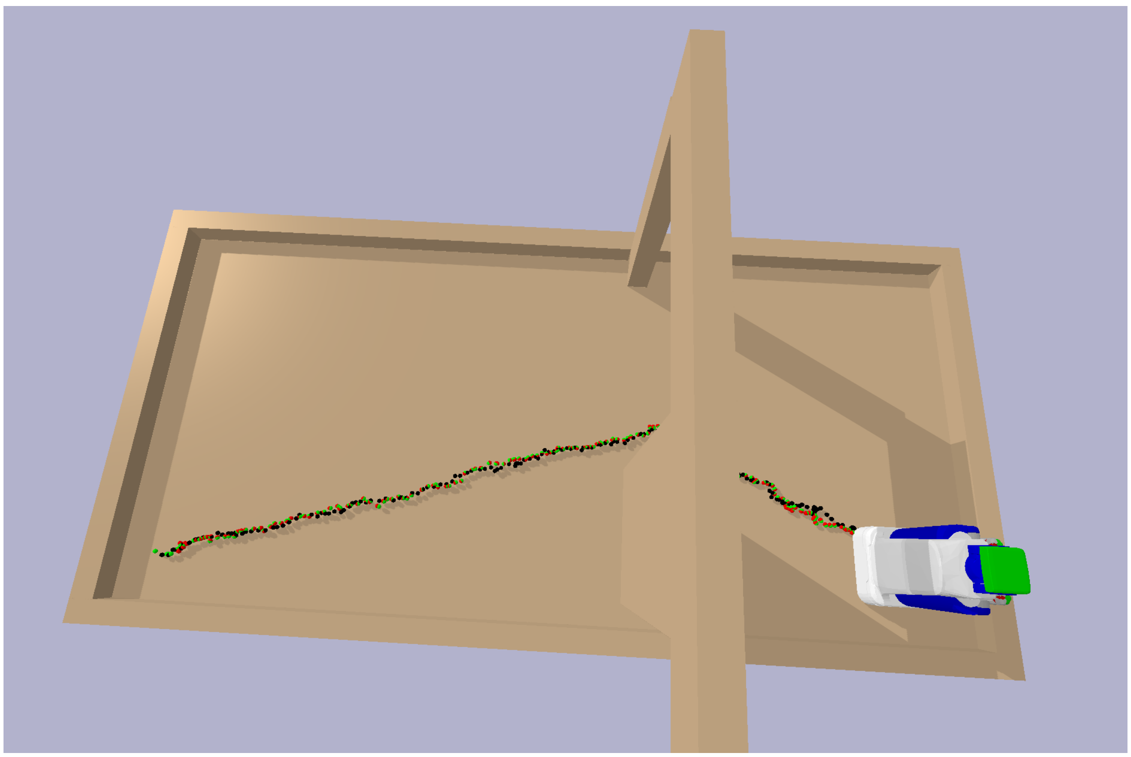

Figure 1: PyBullet simulation environment with the PR2 robot and static obstacles.

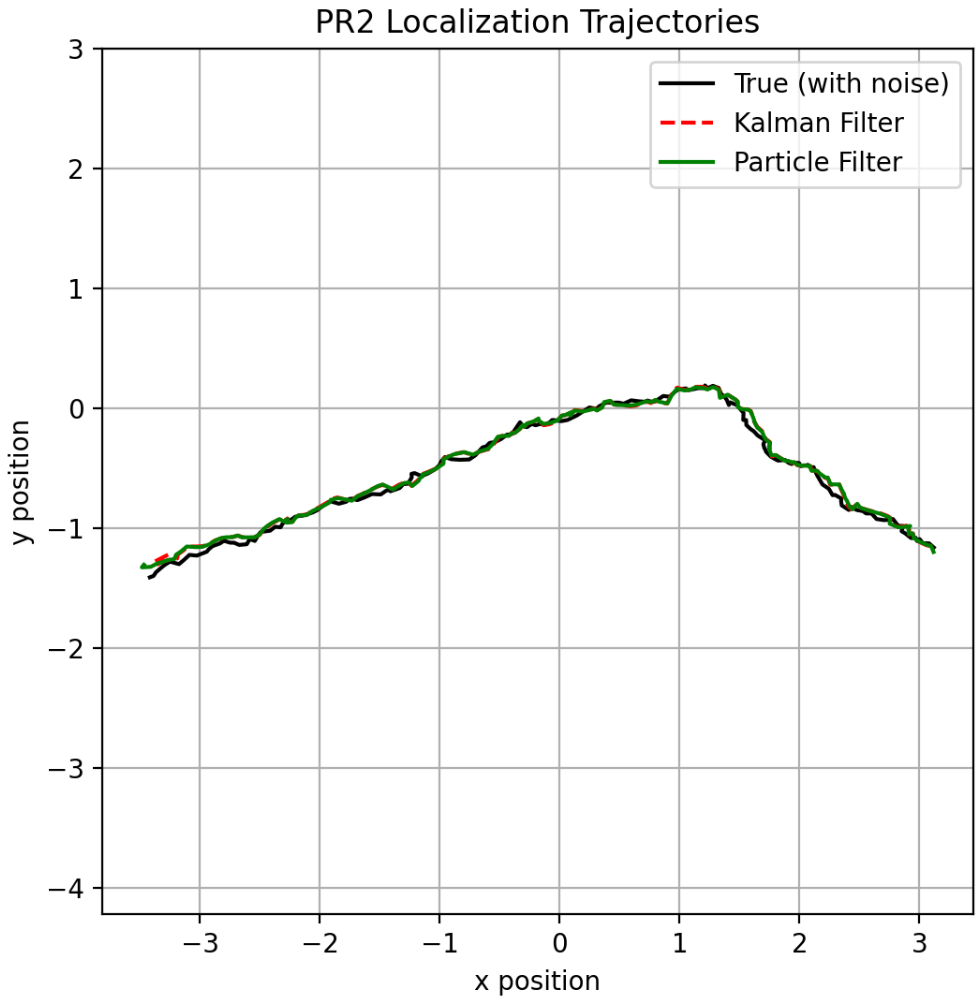


Figure 2: Predefined two-segment trajectory of the PR2 robot, showing the change in direction.

During the simulation:

- The noisy true robot state is visualized in black.
- The Kalman filter estimate is shown in red.
- The particle filter estimate is shown in green.

This visualization allows direct qualitative comparison between the two estimation methods under identical motion and sensing conditions.

# 3 RESULTS

This section presents the experimental results for the PR2 localization task and compares the performance of the Kalman filter (KF) and the particle filter (PF) under different noise conditions.

Table 1: Localization Performance under Different Noise Levels

| Motion Noise | Sensor Noise | RMSE KF | RMSE PF | Mean NEES (KF) |
|---|---|---|---|---|
| 0.01 | 0.04 | 0.0300 | 0.0301 | 2.37 |
| 0.02 | 0.08 | 0.0598 | 0.0579 | 2.30 |
| 0.04 | 0.16 | 0.1289 | 0.1086 | 2.32 |
| 0.08 | 0.32 | 0.2631 | 0.2103 | 2.49 |

All experiments were conducted in a PyBullet simulation environment with a fixed predefined trajectory, while the motion and sensor noise levels were systematically varied.

### 3.1 Experimental Setup

The PR2 robot was commanded to follow a two-segment planar trajectory with a change in direction midway through the motion, introducing moderate nonlinearity into the localization problem. Four different noise configurations were tested, ranging from low to extreme noise levels, by scaling both the motion noise and the sensor noise simultaneously. For each configuration, the same trajectory and initial conditions were used to ensure a fair comparison between the two filters.

Localization accuracy was evaluated using the root mean square error (RMSE) between the estimated and true positions. In addition, the normalized estimation error squared (NEES) was computed for the Kalman filter to assess estimation consistency. Trajectory plots and error-versus-time curves were also generated to provide qualitative insight into the filters' behavior.

Table 1 summarizes the quantitative localization performance of the Kalman filter and particle filter under different motion and sensor noise levels.

### 3.2 Discussion

Overall, the experimental results demonstrate that while the Kalman filter performs well under low-noise, near-linear conditions, its accuracy and consistency degrade as uncertainty increases. The particle filter, although computationally more expensive, provides superior robustness under higher noise levels by maintaining a more flexible representation of the robot's belief state.

These findings are consistent with the theoretical properties of the two methods and confirm that particle filters are better suited for localization problems involving significant uncertainty or deviations from linear-Gaussian assumptions.

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## References

[1] S. Thrun, W. Burgard, and D. Fox, "Probabilistic Robotics," MIT Press, Cambridge, MA, USA, 2005.

[2] H. Durrant-Whyte and T. Bailey, "Simultaneous Localization and Mapping: Part I," *IEEE Robotics & Automation Magazine*, vol. 13, no. 2, pp. 99–110, 2006.

[3] R. E. Kalman, "A New Approach to Linear Filtering and Prediction Problems," *Transactions of the ASME–Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

[4] Y. Bar-Shalom, X. R. Li, and T. Kirubarajan, "Estimation with Applications to Tracking and Navigation," John Wiley & Sons, 2001.

[5] A. Doucet, N. de Freitas, and N. Gordon, "Sequential Monte Carlo Methods in Practice," Springer, New York, NY, USA, 2001.

[6] S. Thrun, "Particle Filters in Robotics," in *Proc. of the 17th Annual Conference on Uncertainty in AI (UAI)*, pp. 511–518, 2002.

[7] F. Dellaert, D. Fox, W. Burgard, and S. Thrun, "Monte Carlo Localization for Mobile Robots," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, pp. 1322–1328, 1999.

[8] D. Fox, W. Burgard, and S. Thrun, "Markov Localization for Mobile Robots in Dynamic Environments," *Journal of Artificial Intelligence Research*, vol. 11, pp. 391–427, 1999.